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# A GAN-based Data Augmentation Method for Imbalanced Multi-class Skin Lesion Classification

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**ABSTRACT** Skin cancer is one of the most common types of cancer globally. Despite the remarkable advancements of deep learning methods in computer vision, automatic diagnosis of skin diseases still faces challenges such as limited data and class imbalance. Generative Adversarial Networks (GANs), which can synthesize realistic data, appear as an alternative to mitigate these issues. However, for imbalanced data, unconditional GANs either generate uneven data distribution or neglect universal knowledge of the whole dataset, while state-of-the-art (SOTA) conditional GANs suffer from performance degradation due to the mode collapse of minority classes. This paper proposes a two-stage GAN-based method to synthesize fine-grained and diverse 256×256 pixels skin lesion images for the imbalanced dataset, named Self-Transfer GAN (STGAN). STGAN first learns universal knowledge from all classes then transfers this shared knowledge to each class and fuses it with class-specific knowledge to synthesize high-quality images. Furthermore, based on STGAN, a framework to enhance the classification performance is established. Both data generation and classification tasks are evaluated on HAM10000 dataset. In terms of Fréchet Inception Distance (FID), Inception Score, Precision, and Recall, STGAN improved by 16%, 16%, 4%, 33% compared with SOTA conditional StyleGAN2. For classification, the STGAN-based framework achieves remarkable results, with an Accuracy of 98.23%, Sensitivity of 88.85%, Precision of 90.23%, F1-score of 89.48%, and Specificity of 98.34%.

**INDEX TERMS** Data augmentation, Data imbalance, Generative adversarial networks, Skin lesion classification.

## I. INTRODUCTION

Skin cancer has become one of the most prevalent types of cancer worldwide, with approximately 9500 cases in the United States every day [1]. It encompasses various types, including basal cell carcinoma, squamous cell carcinoma, melanoma, etc. [2]. Among these, melanoma is the most dangerous and accounts for the highest number of deaths [3]. However, studies have demonstrated that skin cancer can be effectively treated when diagnosed at early stage and managed appropriately [3], [4]. Therefore, early detection is crucial for improving the survival rate of skin cancer patients.

The most important aspect of the clinical diagnosis and treatment of skin cancer lines in the observation of skin lesions. Dermoscopy, a non-invasive imaging technique, provides rich details and accurate color representation of skin lesions, making it widely applied in the diagnosis of skin diseases. With the use of dermoscopy imaging, the diagnostic accuracy can be boosted to 75%-84% [5]. However, the visual examination of dermoscopy images by dermatologists is commonly perceived as a challenging and time-consuming task, often subject to errors and subjectivity [6]. Moreover, for patients residing in remote areas, it is hard to access specialized dermatologists. Hence, an automated computer-aided diagnostic system is highly demanded.

Despite the significant advancements achieved by recent deep learning methods in visual data, they still face following challenges in automated skin disease diagnosis. First, the available annotated data are severely limited, as image acquisition and labeling require specialized equipment and trained medical professionals. Taking the publicly available



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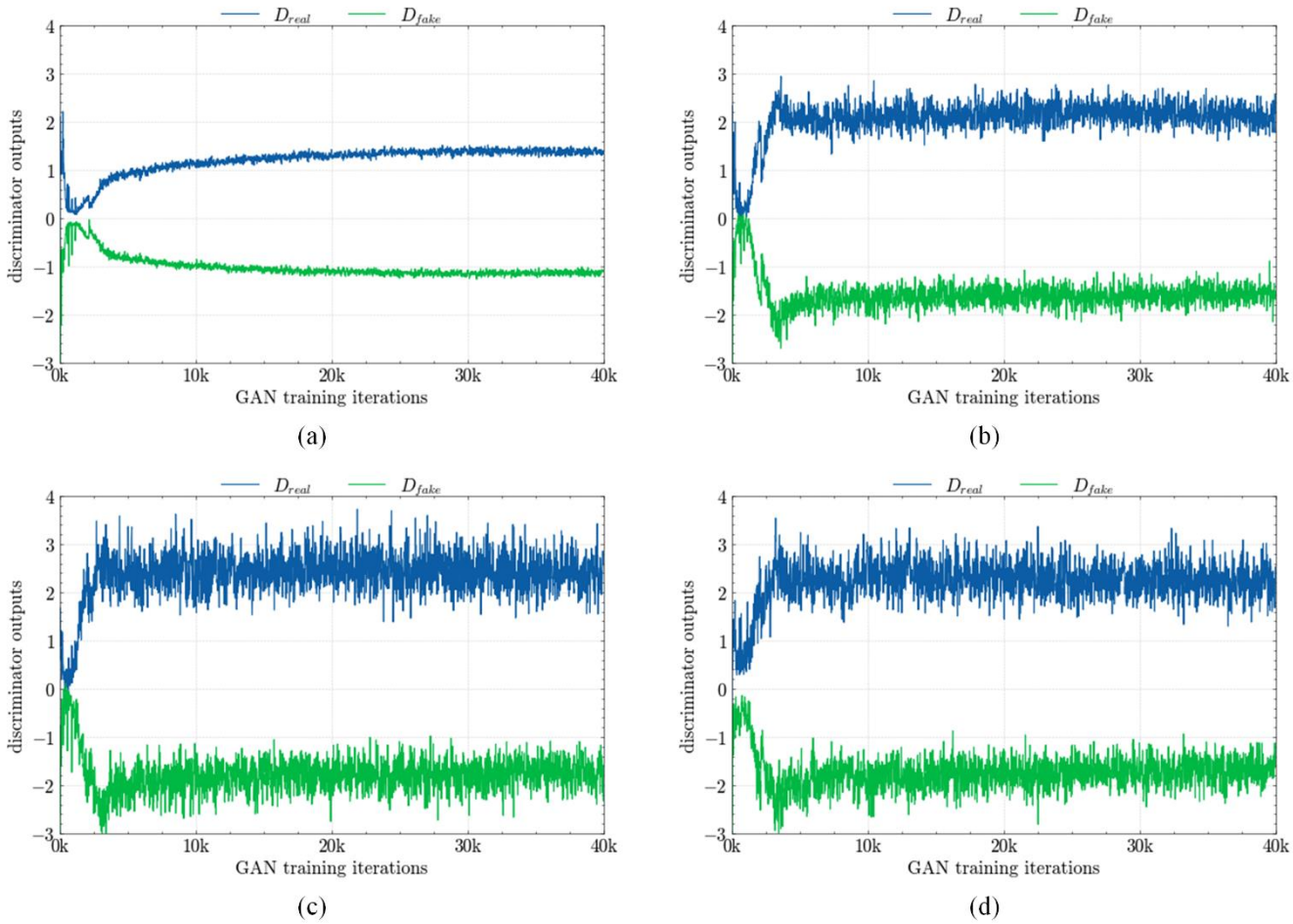



**FIGURE 1.** **Progression of discriminator outputs during training for 4 classes from HAM10000 dataset. Each vertical slice shows the raw logits output by discriminator for real data (blue line) and generated data (green line). (a) NV. (b) AKIEC. (c) DF. (d) VASC**

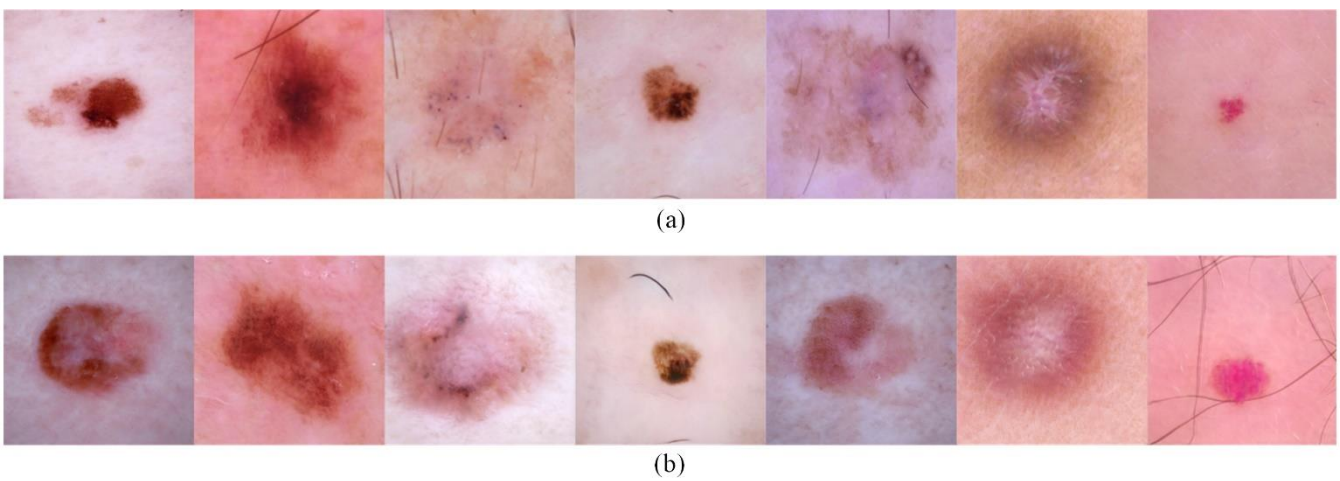



**FIGURE 2.** **Real Images and generated images by STGAN. (a) Real images. (b) Generated images by STGAN.**

HAM10000 [7] dataset as an example, it consists of 10,015 images, while the widely used benchmark dataset in computer vision, like ImageNet [8], contains around 1.3 million training images. Besides, due to variations in the prevalence of different skin diseases, skin disease datasets often suffer from class imbalance issues. Class imbalance can arise from either inter-class imbalance or intra-class imbalance [9]. Inter-class imbalance refers to situations where the number of samples in some classes is significantly smaller than in others, it can be quantified by imbalance ratio,



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which is calculated as the ratio between the number of samples in the largest class to the number of samples in the smallest class [10]. The imbalance ratio of the HAM10000 dataset is 58.3. The class with the largest sample size, melanocytic nevus (NV), has 6705 images, while the class with the smallest sample size, dermatofibroma (DF), only has 115 images. Traditional Convolutional Neural Networks (CNN) classifier architectures are typically designed to work well for balanced datasets, they tend to treat minority classes as rare occurrences, or even as noise, when confronted with inter-class imbalance [11]. Intra-class imbalance refers to the attribute bias within a class, it can be considered as a form of inter-class imbalance in the context of fine-grained visual categorization. Intra-class imbalance can also negatively affect the performance of CNN classifiers. The presence of intra-class imbalance is closely intertwined with the "small disjuncts problem" [12], [13], a problem that has been demonstrated to significantly impair classification performance [14]. Small disjuncts refers to classification rules that cover only a limited number of training samples, and the scarcity of samples renders rule induction more susceptible to errors [15], [16].

Traditional Data Augmentation (TDA) techniques, such as color space transformations (e.g., adjusting saturation, contrast, brightness, and hue of the training image), geometric transformations (e.g., flipping, cropping, rotation, translation), mixing images, and random erasing, can effectively inflate the size of training dataset, thereby mitigating the issues of limited labeled data and class imbalance problem [17]. Geometric and color space transformations are the most commonly employed data augmentation techniques in skin disease classification tasks [18]. Perez, *et al.* [19] investigated 13 TDA scenarios for skin lesion classification trained on CNN classifiers, their experiments revealed that color space transformations and geometric transformations could enhance the classification performance. However, image mixing and random erasing did not yield improvements, this is attributed to the fact that these two data augmentation techniques cannot guarantee that the transformed images retain the relevant skin lesion features of the source images. The works in [20], [21] also reported that the application of geometric transformations and color space augmentations could improve the performance of skin lesion classification. Although TDA techniques, like geometric and color space transformations, can improve skin lesion classification performance, it faces some drawbacks. First, the transformed images with small modifications have a similar distribution to the original images and do not bring any new visual features to the training data, resulting in limited improvement in performance [22], [23]. Additionally, the associated distortion magnitude parameter of geometric and color space transformations need to be carefully chosen, since heavily transformed images could alter the semantic meaning of the medical images [17], [24]. Moreover, for imbalanced dataset, naïve augmentive oversampling with basic image transformations could cause overfitting on the minority class [17].

In contrast to TDA, Generative Adversarial Networks (GANs) [25] aim to learn the underlying data distribution and generate realistic looking images. Recent advancements in GANs indicate that their ability to represent complex and high-dimensional data can serve as an advanced data augmentation method to increase the amount of training samples and mitigate the imbalance problem of skin lesion dataset [22], [26], [27], [28], [29], [30]. Basically, there are two GAN-based solutions for data augmentation, one is unconditional GANs and the other is conditional GANs [31], [32], [33]. As for unconditional GANs, there are two training regimes. One regime involves training a single unconditional GAN model using the entire training set. A drawback of this training regime is that when the training data is imbalanced, the distribution of generated data between different classes remains imbalanced as well [34], leading to exacerbation of the class imbalance issue in downstream classification tasks; The other regime is to train an unconditional GAN model for each class separately. However, this regime fails to utilize global knowledge or information that may be available across all classes.

On the contrary, conditional GANs incorporate class labels during the training process, enables it to explore the universal knowledge and class specific knowledge in one model, making it a natural choice for multi-class image generation. However, findings in [35] reveal that despite achieving impressive results on metrics, samples generated by conditional GANs lack diversity. Moreover, it has been recently found that, state-of-the-art (SOTA) conditional methods like BigGAN [36] and StyleGAN2 [37] with augmentation, suffer performance degradation due to the mode collapse of minority classes [38]. Furthermore, the performance of conditional GANs degrade more in comparison with unconditional GANs in imbalanced scenario [39].

For investigating the mode collapse phenomenon of conditional GANs on minority classes further, we train the recently SOTA conditional StyleGAN2 (cSG2) with Adaptive Discriminator Augmentation (ADA) [40] on HAM10000 dataset, and take a closer look at the outputs of the discriminator for each class. Fig. 1 shows the outputs distributions of the discriminator on the majority class melanocytic nevus (NV) and three minority classes: actinic keratosis (AKIEC), dermatofibroma (DF), and vascular lesion (VASC). During the training progress, for NV, the discriminator outputs of real images and generated images keep diverging from each other slowly and steadily. In contrast, the discriminator's outputs for the three minority classes, both for real images and generated images, exhibits substantial fluctuations around some value after approximately 2.5k iterations. This suggests that the model fails to acquire class-specific knowledge for these three



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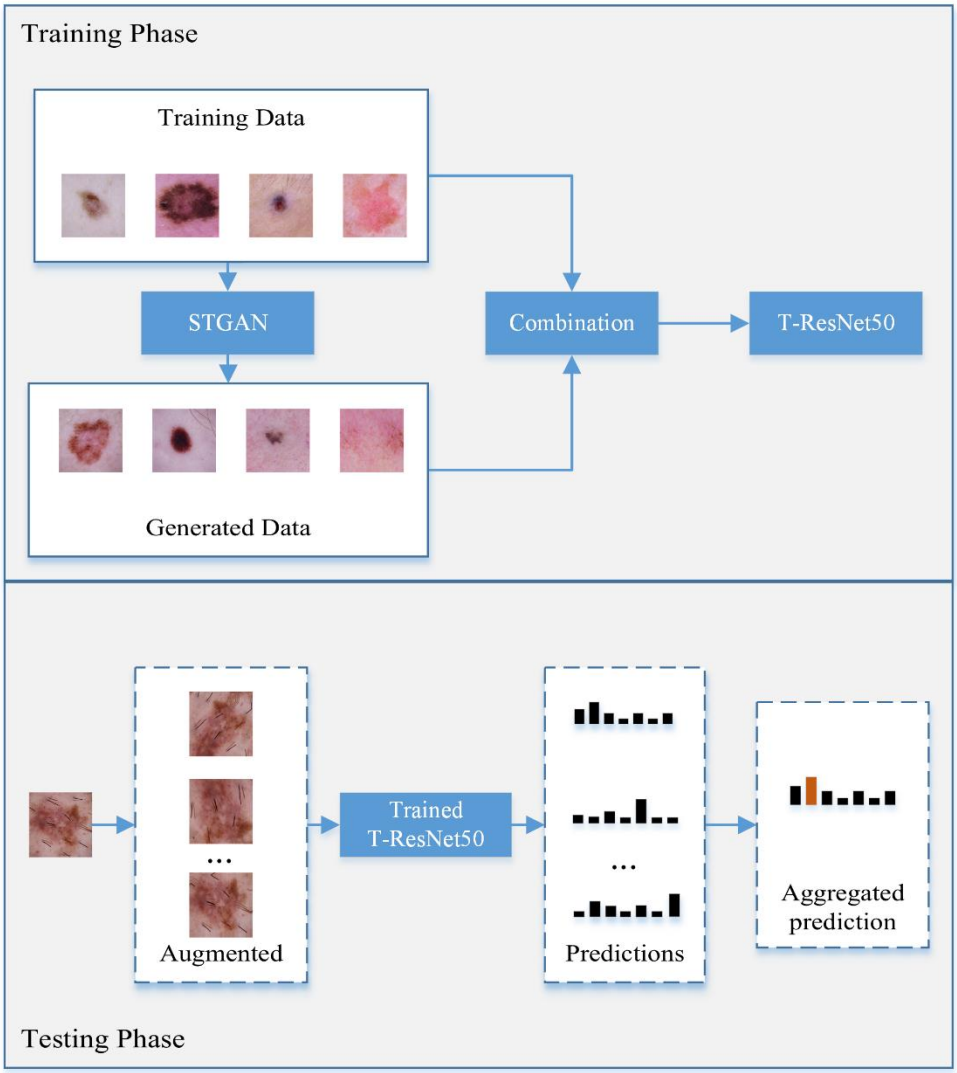



**FIGURE 3.** **An overview of the proposed STGAN-based framework for improving CNN-based classification results. STGAN is utilized to generate synthetic images and then the generated images and training set is combined to further train T-ResNet50. In the testing phase, TTA is employed. For a given test image, various data augmentation methods are applied, and each augmented version is fed into the trained T-ResNet50 model. The final prediction is then aggregated from the outputs corresponding to these augmented images.**

minority classes and only learns global knowledge through the shared generator parameters. This is further evidenced by the high precison and low recall (a.k.a mode collapse) of minority classes in Table II.

Based on the reasons discussed above, a two-stage GAN-based method for skin lesion image synthesis, named STGAN, is proposed in this study. STGAN decouples the learning of universal knowledge and class-specific knowledge to mitigate the mode collapse of minority classes. Specifically, an unconditional GAN is first trained using all samples to learn the shared universal knowledge of skin lesion images, and then the learned universal knowledge is propagated to each class and fused with class-specific knowledge to generate images for each class in second stage. Furthermore, given the very limited number of samples for some classes in the second stage, self-supervised learning [41] and Freeze-D [42] are introduced in the discriminator to enhance sample efficiency.

In contrast to conditional GANs, STGAN builds upon the knowledge transfer and separates the learning of global and class-specific knowledge. This separation offers two advantages. First, the global knowledge acquired by the model in the first stage facilitates efficient learning of class-specific knowledge. Second, this separation allows for better control over the learning of class-specific knowledge, thereby mitigating mode collapse of the minority classes.

Furthermore, based on STGAN, a whole framework to enhance the skin lesion diagnosis performance for limited and imbalanced data is established (Fig. 3). First, STGAN is utilized to generate diverse and visual-appealing images, then the generated data and training data are combined to train a ResNet50 [43] model, which is initialized with pre-trained



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weights on ImageNet. In the testing phase, Test-Time Augmentation technique (TTA) [19], [44], [45] is employed to further boost the classification results.

In summary, the following contributions are proposed:

1. A two-stage GAN-based image synthesis method is proposed for generating diverse and visually-appealing dermoscopy images for both majority and minority classes within an imbalanced skin lesion dataset. (Fig. 2(b)).
2. Images generated by STGAN contain high information gain, which can improve the performance of CNN-based classifier for automated skin disease diagnosis.
3. A comprehensive framework for processing dermoscopy images is developed, encompassing image generation, classifier construction, and image classification. This framework can provide remarkable classification results on limited and imbalanced datasets.

The remainder of this paper is arranged as follows: in Section II, related work is reviewed, followed by the presentation of our methodology in Section III. Experimental results are presented and analyzed in Section IV, and the conclusion is provided in Section V.

## II. RELATD WORK

In this section, the review and discussion of previous research closely related to this study are conducted. The related works are categorized into three main themes: imbalanced data, GAN transfer learning and GANs in skin lesion image synthesis.

### A. IMBALANCED DATA

Data imbalance refers to a situation in a dataset where one or more classes have a much larger samples compared to other classes. Numerous efforts have been devoted to developing classification models for this uneven data distribution [13], [46], [47], [48]. However, only a limited number of studies have focused on learning generative models for imbalanced data distribution. Rangwani, *et al.* [38] proposed a group Spectral Regularizer (gSR) for conditional GANs to alleviate mode collapse. Rangwani, *et al.* [49] introduced continuity in class label embedding vectors to improve the quality of generated images for minority classes. However, these techniques are primarily designed for conditional GANs. Rangwani, *et al.* [39] found that the performance of conditional GANs deteriorates more in the presence of data imbalance compared to unconditional GANs and proposed the use of a pre-trained classifier to guide an unconditional GAN in generating high-quality images for each class. Nonetheless, the prerequisites for a well-pretrained classifier in [39] could potentially impose limitations.

### B. GAN TRANSFER LEARNING

Transfer learning is a machine learning technique that utilizes pre-trained parameters from other dataset as the initial parameters, rather than random initialization. By adopting this approach, knowledge can be transferred from one dataset (i.e., source domain) to another (i.e., target domain), resulting in reduced training time and data size requirements. Several researches have explored transfer learning in the context of GANs [42], [50], [51], [52]. Wang, *et al.* [52] is one of the first to explore transfer learning in GANs. Their experimental findings revealed that leveraging knowledge from pre-trained models can lead to reduced training time and enhanced image generation quality in scenarios with limited data. Noguchi and Harada [50] introduced a novel approach that involves keeping all convolutional kernel parameters fixed and only updating shift and scale parameters in the generator. Wang, *et al.* [51] proposed a miner network to identify a narrower region of the prior distribution where images generated by pre-trained model are similar to the target data, so as to ease the following finetuing step on the target data. Mo, *et al.* [42] proposed a simple method, Freeze-D, which fixs the lower layers' parameters in discriminator when fine-tuning GANs.

While GAN-based transfer learning techniques are effective in addressing the limited data problem, their effect on imbalanced dataset have not been well explored. Different from GAN-based transfer learning, the proposed method STGAN address the problem of imbalanced learning in GANs. Moreover, despite both being built upon the concept of knowledge transfer, STGAN involves transferring shared knowledge between different classes to each individual class in one domain, whereas current GAN-based transfer learning focuses on transferring knowledge between different domains.

### C. GAN IN SKIN LESION IMAGE SYNTHESIS

Yi, *et al.* [30] utilized categorical generative adversarial network (catGAN) [53] with Wasserstein distance [54] for dermoscopy images feature learning. They generate 64×64 pixels images and achieve an average precision score of 0.424 with only 140 labeled images from the International Skin Imaging Collaboration (ISIC) 2016 Challenge dataset [55]. Shoohi and Saud [56] proposed using unconditional DCGAN [57] to synthesis 64×64 pixels samples for the minority class for imbalanced Malaria dataset. However, skin lesion images should be generated in a high-resolution (e.g., 256×256 pixels) as it enables the identification of malignancy markers that differentiate between benign and malignant lesions with higher accuracy.

With the goal of synthesizing high-quality skin lesion images, Baur, *et al.* [58] proposed an unconditional deeply discriminated GAN (DDGAN) to generate 256×256 pixels skin lesion image. The effectiveness of DDGAN was compared with two other popular GAN architectures, namely the DCGAN [57] and LAPGA [59]. The results showed that both the DDGAN and LAPGAN are capable of learning the distribution of the datasets and synthesizing realistic samples. However, in the case of imbalanced data, the samples generated by DDGAN remains class imbalance as well, making it unsuitable as a data augmentation method for



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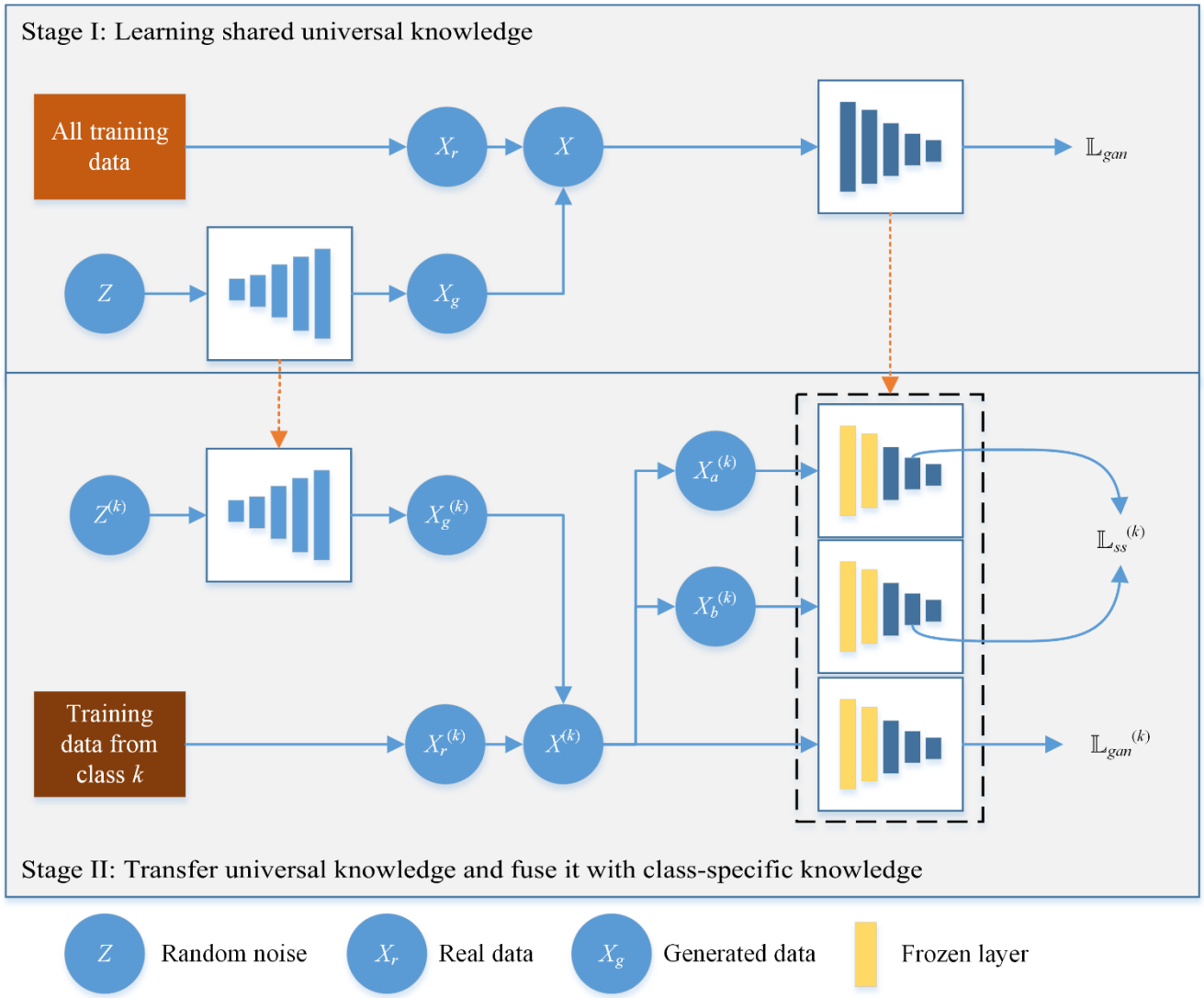



**FIGURE 4. An overview of STGAN. STGAN is a two-stage image generation method based on GAN transfer learning. In the first stage, an unconditional GAN is used to learn the shared universal knowledge of skin lesion images. This pre-trained unconditional GAN serves as the initialization for the second-stage, which continues training on images from each class. Meanwhile, in the second stage, STGAN employs Barlow Twins to improve sample efficiency and Freeze-D to enhance knowledge transfer efficiency.**

downstream classification tasks. Qin, *et al.* [22] proposed SL-StyleGAN to generate 256×256 pixels skin lesion images with modified style controlling method and simplified the generator and discriminator architectures of StyleGAN [60]. However, SL-StyleGAN was trained on individual class data thus did not make the use of universal knowledge of the whole dataset.

Abdelhalim, *et al.* [26] proposed SPGGAN-TTUR, they utilized conditional Progressive Growing of GAN(cPGGAN) [61] with a self-attention mechanism to generate 256×256 pixels skin lesion images. However, SPGGAN-TTUR ignored the class imbalance problem in skin lesion dataset, which can lead to a degradation in the quality of the generated images for minority classes.

Different from these works, STGAN avoids the performance degradation issue faced by GANs on imbalanced datasets. It can generate high resolution realistic images with fine-grained detail for each class, thereby enhancing the performance of downstream CNN classifiers for skin lesion diagnosis.

## III. METHODOLOGY

In this section, we introduce our method aim at improving the CNN-based classification results of skin lesion image by generating high-quality images to mitigate the impact of limited labeled data and the imbalance between data classes (see Fig. 3). The framework of the proposed STGAN is shown in Fig. 4. STGAN first employs an unconditional GAN model to learn universal knowledge of all training samples (subsection III-A), and then transfer this shared knowledge to each class and fuse it with class-specific knowledge for high-quality image generation (subsection III-



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B). Finally, images generated by STGAN are combined with real images to further train a ResNet50 model (subsection III-C).

### A. UNSUPERVISED LEARNING OF UNIVERSAL KNOWLEDGE

STGAN first employs an unconditional GAN to learn the global knowledge of the entire dataset. The unconditional GAN framework consists two adversarial neural networks: the generator $G$ and the discriminator $D$. The generator $G$ parameterized by $\theta$ takes random noise vector $Z$ as input to generate sample $X_g = G_\theta(Z)$. On the other hand, the discriminator $D$ functions like a classifer, it takes real data $X_r$ and the generated data $X_g$ as input and tries to optimize its parameters $\phi$ by learning to distinguish between these two types of data.

The core idea of GAN lies in the interplay between the generator and the discriminator: the generator attempts to produce realistic data to deceive the discriminator, while the discriminator strives to enhance its ability to distinguish real data from generated data. This adversarial process drives the generator to continually improve the quality of generated data, thereby approximating the distribution of real data.

The loss function of discriminator can be defined as:

$$\mathbb{L}_D = -\mathbb{E}_{X_r \sim p_d}[\log D_\phi(X_r)] - \mathbb{E}_{Z \sim p_Z}[\log(1 - D_\phi(G_\theta(Z)))]. \quad (1)$$

The loss function of generator can be defined as:

$$\mathbb{L}_G = \mathbb{E}_{Z \sim p_Z}[\log(1 - D_\phi(G_\theta(Z)))]. \quad (2)$$

And the complete two-player minimax game can be formulated as

$$\min_\theta \max_\phi \left\{ \mathbb{L}_{gan} = \mathbb{E}_{X_r \sim p_d}[\log D_\phi(X_r)] + \mathbb{E}_{Z \sim p_Z}[\log(1 - D_\phi(G_\theta(Z)))] \right\} \quad (3)$$

where, $p_d$ is the true data distribution, and $p_Z$ is the random noise distribution.

### B. KNOWLEDGE TRANSFER AND FUSION

The discriminator and generator trained in the first stage serve as the initialization for the second stage, where they are further trained on data from each individual class to learn class-specific knowledge. However, the binary classifiication task in this stage could be too easy for the discriminator to gain substantial discriminative capacity and providing sufficient feedback to train the generator. Thus, a more challenging task is assigned to the discriminator by employing self-supervised learning in the second stage to prevent overfitting. Furthermore, Freeze-D is also adopted to enhance training stability and improve knowledge transfer performance.

#### 1) SELF-SUPERVISED LEARNING

Recent researches [62], [63], [64] in self-supervised learning have shown that maximizing the agreement of different views (where views refer to same image go through different data augmentations) can facilitate the learning of informative and rich semantic feature representations by model. Drawing inspiration from this insight, STGAN integrate self-supervised learning into the discriminator in second stage. The advantage of employ self-supervised learning is twofold. First, if the discriminator is able to learn semantically-rich features, it will also be better at distinguishing real data from generated data, thereby prompting the generator to produce high-quality images. Second, enforcing the discriminator to learn high-quality features elevates the difficulty of its learning task, thereby preventing the discriminator from memorizing data in very limited samples and mitigating the risk of overfitting. By applying self-supervised learning, STGAN can effectively leverage the limited samples in underrepresented classes, making the most out of the available data and improving the performance in handling imbalanced dataset.

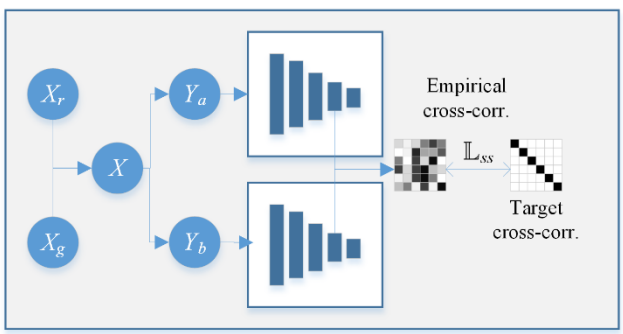



**FIGURE 5. The framework of Barlow Twins. Two different data augmentation methods are applied to both real samples and generated samples. The two feature vectors extracted by the penultimate layer of the discriminator are used to compute the cross-correlation matrix (cross-corr.), which is then compared with the identity matrix (target cross-corr.) to calculate the loss.**

Specifically, Barlow Twins [41] is intergrated into the discriminator (Fig. 5), as it offers simplicity and does not require a large batch size. The core idea of Barlow Twins is to ensure maximam similarity and minimal information redundancy between feature vectors obtained from different views through a shared feature extractor. For a given batch input $X^{(k)}$ from class $k$, two views $X_a^{(k)}$ and $X_b^{(k)}$ are obtaind through two different data augmentations. These two different views are further input to the discriminator to get feature representations $Y_a^{(k)}$ and $Y_b^{(k)}$ (output of the penultimate layer of the discriminator). Note that we assume $Y_a^{(k)}$ and $Y_b^{(k)}$ are mean-centered along the batch dimension for notational convenience. These two feature representations are then used to calculate the empirical cross-correlation matrix given as:

$$\mathbf{C}_{ij}^{(k)} \triangleq \frac{\sum_{(y_a^{(k)}, y_b^{(k)}) \in (Y_a^{(k)}, Y_b^{(k)})} y_{a,i}^{(k)} y_{b,j}^{(k)}}{\sqrt{\sum_{y_a^{(k)} \in Y_a^{(k)}} \left(y_{a,i}^{(k)}\right)^2} \sqrt{\sum_{y_b^{(k)} \in Y_b^{(k)}} \left(y_{b,j}^{(k)}\right)^2}} \quad (4)$$



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where $i$, $j$ index the vector dimension. The loss is given as:

$$\mathbb{L}_{ss}^{(k)} = \sum_{i}(1-\mathbf{C}_{ii}^{(k)})^2 + \lambda\sum_{i}\sum_{j\neq i}(\mathbf{C}_{ij}^{(k)})^2, \quad (5)$$

where $\lambda$ is a scaling factor.

The final training loss function of STGAN in stage two is:

$$\mathbb{L}_{GAN}^{(k)} = \mathbb{L}_{gan}^{(k)} + \lambda_{ss}\mathbb{L}_{ss}^{(k)}, \quad (6)$$

where $\lambda_{ss}$ is a balancing hyperparameter.

#### 2) FREEZE-D

Freeze-D [42] is employed to stabilize the training process in the second stage and enhance the performance of knowledge transfer. Freeze-D achieves this by freezing the top $n$ highest-resolution layers of the discriminator during the transfer process, which has recently demonstrated strong results. One thing should be noted that, Freeze-D was originally proposed for knowledge transfer between different domains, whereas STGAN utilize it for transfer universal knowledge to class-specific knowledge in one domain.

### C. STGAN-BASED SKIN LESION CLASSIFICATION FRAMEWORK

Fig. 3 shows an overview of the proposed STGAN-based automated multi-class imbalanced skin lesion dermoscopy image classification framework, where STGAN is employed as a data augmentation method to improve the classification results.

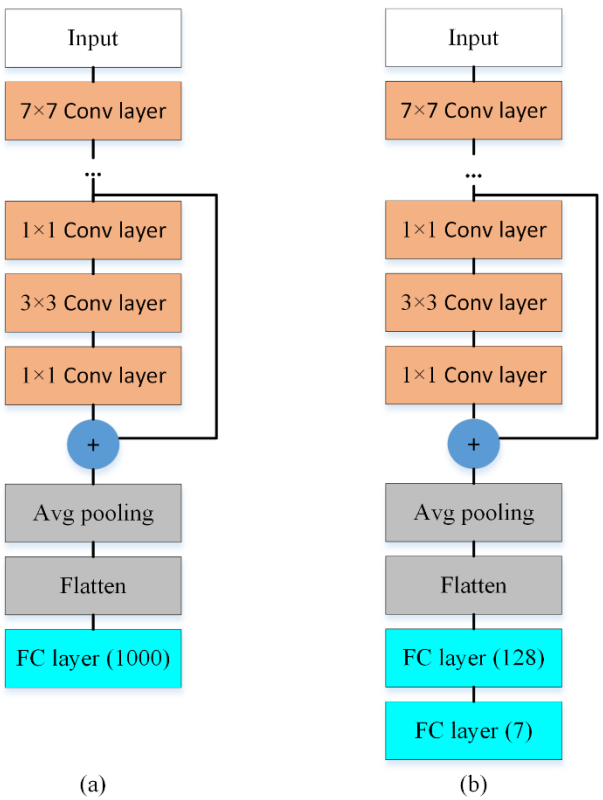



**FIGURE 6. Architecture of CNN classifier. (a) ResNet50. (b) T-ResNet50.**

In this work, 50-layer ResNet (Deep Residual Networks) [43] is chosen as a CNN classifier. ResNet first introduced residual connections in DNNs, which addresses issues such as gradient vanishing and gradient explosion encountered in DNNs. This enables the training of very deep neural networks without compromising performance and has demenstraed excellent performance on various computer vision tasks, including skin lesion classification [65], [66], [67], [68]. Instead of training ResNet50 from scratch, we utilize transfer learning and initialize ResNet50 with ImageNet pre-trained weights (T-ResNet50). Additionally, the original single fully connected layer in ResNet50 is replaced with two fully connected layers. The first fully connected layer maps the 2048-dimensional output from the flatten layer to a 128-dimensional vector, while the second fully connected layer maps this 128-dimensional vector to 7 logits corresponding to the 7 classes in the HAM10000 dataset (see Fig. 6(b)).

As shown in Fig. 3, the training set is used to train the STGAN, and then the synthetic samples generated by STGAN are combined with the training set to further train the T-ResNet50 model. In the testing phase, instead of presenting the trained T-ResNet50 with the origin testing image, TTA is employed to further boost the classification performance. Concretely, we repeatedly introduce the TDA augmented images to the model, and compute the average predictions for each corresponding image and utilize this aggregated prediction as the ultimate inference.

## IV. EXPERIMENTS AND RESULTS

Both image generation and classification experiments are conducted to assess the effectiveness of the proposed method. The recent state-of-the-art cSG2 is chosen as the baseline. In subsection IV-E, quantitative and qualitative studies of STGAN and the baseline in terms of image generation performance are provided. In subsection IV-F, the performance of the STGAN-based framework on skin lesion image classification is investigated.

### A. DATASET

TABLE I THE CLASS DISTRIBUTION FOR HAM10000 DATASET

| Category | Total | Train | Test |
|---|---|---|---|
| MEL | 1113 | 890 | 223 |
| NV | 6705 | 5364 | 1341 |
| BCC | 514 | 411 | 103 |
| AKIEC | 327 | 262 | 65 |
| BKL | 1099 | 879 | 220 |
| DF | 115 | 92 | 23 |
| VASC | 142 | 114 | 28 |

The proposed method is evaluated on HAM10000 dataset, which is a publicly available dataset. The dataset is heavily imbalanced. It consists of 10,015 dermoscopic skin lesion images from seven skin diseases with a resolution of $600\times450$ pixels. For fair comparison, this study follows



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[22], [27], [69], [70], [71] and split the HAM10000 dataset into a training and test set by the proportion of 8:2, each containing 8012 images and 2003 images, respectively (Table I). These works are followed based on two reasons. First, similar to this research, the works in [22], [27] also employed GAN-based data augmentation method to mitigate limited data and class imbalance problem. Second, very recent works in [69], [70], [71] have demonstrated notable performance in the field of skin lesion classification.

### B. EVALUATION METRICS OF GAN

In this work, following metrics are used for quantitative assessment of the performance introduced by STGAN in image generation.

#### 1) FRÉCHET INCEPTION DISTANCE (FID)

FID is the 2-Wasserstein distance between the real image distribution $p_r$ and the generated image distribution $p_g$. It measures the similarity between these two distributions to access the quality of the generated images and is given by:

$$\begin{aligned}\mathrm{FID}(p_r, p_g) = &\left\| \mu_{p_r} - \mu_{p_g} \right\|_2^2 \\ &+ Trace\left( C_{p_r} + C_{p_g} - 2\left( C_{p_r} C_{p_g} \right)^{1/2} \right),\end{aligned} \tag{7}$$

where $\{\mu_{p_r}, C_{p_r}\}, \{\mu_{p_g}, C_{p_g}\}$ are the mean and covariance of the final feature vectors extracted by Inception Network [45] from real image and generated image, repectively. A lower FID score indicates that the generated images are more similar to the real images in terms of their high-level features.

#### 2) INCEPTION SCORE (IS)

The Inception Score (IS) [72] is another commonly used metric for GAN evaluation. It is given by:

$$\mathrm{IS} = \exp\left( E_{X_g \sim p_g} [KL(p(y|\ X_g) \| p(y))] \right), \tag{8}$$

where $X_g$ is a sample synthesized by generator $G$, $p(y|\ X_g)$ is the conditional label distribution of sample $X_g$ captured by a pretrained Inception Network on ImageNet, and $p(y)$ is the margin distribution of class label.

The intuition of Inception Score is a high-performance generator should be able to generate varied images (high entropy of $p(y)$) with meaningful objects (low entropy of $p(y|\ X_g)$).

#### 3) PRECISION AND RECALL

This study use improved Precision and Recall [73], as this study aim to mitigate the the mode collapse problem of minority classes and low recall indicates mode collapse [74].

Precision and Recall are estimated from the manifolds in feature space of real data and generated data. Precision quantifies the proportion of generated manifold covered by real data, while Recall denotes the proportion of real data manifold covered by the generator.

Concretely, a pre-trained VGG16 is utilized to extract features from data and create two feature sets: real data feature sets $\mathbf{\Omega}_r$ and generated feature sets $\mathbf{\Omega}_g$. Then a binary function is defined as

$$f(\omega, \mathbf{\Omega}) = \begin{cases} 1, \ \exists \omega' \in \mathbf{\Omega}. \left\| \omega - \omega' \right\|_2 \le \left\| \omega' - \mathrm{N}_k\left( \omega', \mathbf{\Omega} \right) \right\|_2 \\ 0, \ \text{otherwise}, \end{cases} \tag{9}$$

where $\mathbf{\Omega} \in \{\mathbf{\Omega}_r, \mathbf{\Omega}_g\}$, and $\mathrm{N}_k\left(\omega', \mathbf{\Omega}\right)$ is the $k$th nearest feature of $\omega'$ from $\mathbf{\Omega}$. Precision and Recall are defined as

$$Precision\left(\mathbf{\Omega}_r, \mathbf{\Omega}_g\right) = \frac{1}{\left|\mathbf{\Omega}_g\right|} \sum_{\omega_g \in \mathbf{\Omega}_g} f\left(\omega_g, \mathbf{\Omega}_r\right) \tag{10}$$

$$Recall\left(\mathbf{\Omega}_r, \mathbf{\Omega}_g\right) = \frac{1}{\left|\mathbf{\Omega}_r\right|} \sum_{\omega_r \in \mathbf{\Omega}_r} f\left(\omega_r, \mathbf{\Omega}_g\right) \tag{11}$$

### C. EVALUATION METRICS OF CLASSIFICATION

The classification performance is assessed in terms of Accuracy, Sensitivity, Precision, F1-score and Specificity. The definitions of all these metrics are presented below:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{12}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{13}$$

$$Precision = \frac{TP}{TP + FP} \tag{14}$$

$$F1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{15}$$

$$Specificity = \frac{TN}{FP + TN} \tag{16}$$

where TP is true positive, denotes the number of positive samples correctly classified as positive; TN is true negative, denotes the number of negative samples correctly classified as negative; FP is false positive, denotes the number of negative samples misclassified as postive; FN is false negtive, denotes the number of positive samples misclassified as negative.

### D. EXPERIMENTAL SETUP

For all image generation experiments, this study use the official Pytorch [75] implementation of cSG2 as the base framework. Adaptive Discriminator Augmentation (ADA) [40] is also employed in the GAN training. For both the generator and discriminator network, the learning rate is set to 0.0025. The projector network of Barlow Twins in original paper is omitted for simplicity. The original $600 \times 450$ pixels images of HAM10000 dataset are first center-cropped to $450 \times 450$ pixels and then resized to $256 \times 256$ pixels. Fig. 2(a) shows some sample images after this pre-processing.

All classification experiments employ momentum optimizer (SGD) with momentum 0.9, learning rate 0.02 with a consin annealing strategy [76], and a batch size of 128. For both STGAN and cSG2 based augmentation, 18,500 synthesis images are added to the training set, 500 for NV, 3,000 for each of the other categories. Since ResNet50 is employed, both real images and generated images are resized to $224 \times 224$ pixels before training. All experiments are conducted on one NVIDIA Tesla V100 GPU.



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TABLE II EVALUATION RESULTS OF PROPOSED STGAN AND CSG2 IN IMAGE GENERATION

| Class | FID↓ | | IS↑ | | Precision↑ | | Recall↑ | |
|---|---|---|---|---|---|---|---|---|
| | cSG2 | STGAN | cSG2 | STGAN | cSG2 | STGAN | cSG2 | STGAN |
| MEL | 40.35 | 33.57 | 3.12 | 3.53 | 0.71 | 0.76 | 0.22 | 0.29 |
| NV | 19.94 | 17.20 | 2.88 | 2.91 | 0.74 | 0.77 | 0.26 | 0.34 |
| BCC | 47.94 | 41.99 | 2.62 | 3.15 | 0.73 | 0.81 | 0.23 | 0.23 |
| AKIEC | 71.61 | 62.33 | 3.19 | 3.63 | 0.76 | 0.77 | 0.14 | 0.23 |
| BKL | 45.92 | 32.91 | 3.27 | 3.85 | 0.69 | 0.76 | 0.19 | 0.30 |
| DF | 99.26 | 84.02 | 2.73 | 3.49 | 0.83 | 0.83 | 0.13 | 0.15 |
| VASC | 96.91 | 83.67 | 2.32 | 2.74 | 0.84 | 0.83 | 0.06 | 0.19 |
| Average | 60.28 | 50.81 | 2.88 | 3.33 | 0.76 | 0.79 | 0.18 | 0.25 |

↓ means lower is better, ↑ means higher is better

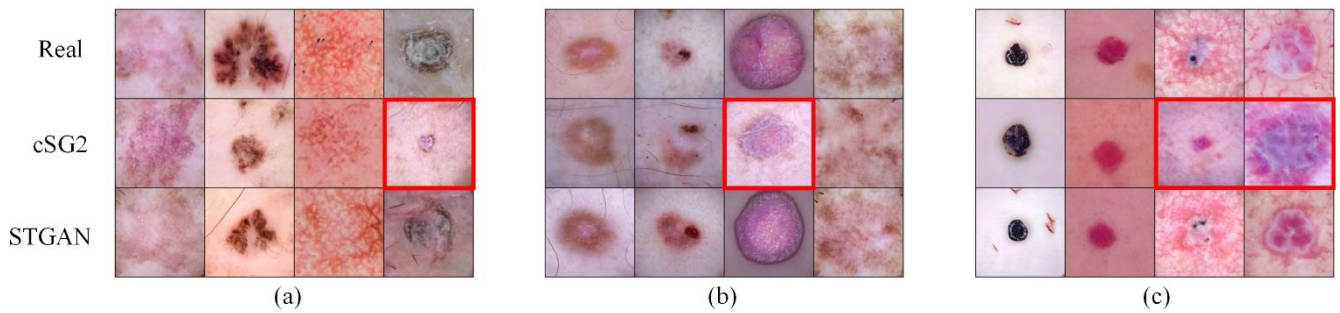



**FIGURE 7. Visual comparison of GANs. The following strategy for image selection is employed. First, four distinct modes are chosen from real images (first row). Then, 1000 images are generated using both cSG2 and STGAN methods and the image that visually resemble real images most closely is selected (second and third rows). (a) AKIEC. (b) DF. (c) VASC.**

### *E. THE PERFORMANCE OF STGAN IN SKIN LESION SYNTHESIS*

#### 1) QUANTITATIVE EVALUATION

Both STGAN and baseline are trained on the training set of HAM10000 introduced before and generated $256 \times 256$ pixels images. They are evaluated quantitatively by the above evaluation metrics. The metrics for each class obtained by STGAN and cSG2 are presented in Table II.

Based on experimental outcomes in Table II, it can be observed that STGAN achieves lower FID (smaller score means a better model) and higher Inception Score across all seven classes compared to cSG2, leading to an improvement of 15.71% in average FID and 15.63% in average Inception Score, indicating that STGAN generates more realistic and diverse semantically meaningful images than baseline.

Further, for cSG2, categories with significantly low training sample sizes, such as AKIEC, DF, and VASC, tend to exhibit high Precision but low Recall. This phenomenon strongly signals the presence of mode collapse. STGAN substantially enhances the Recall of these three categories, while simultaneously maintaining their Precision. This indicates that STGAN can effectively mitigate mode collapse issues for minority classes.

#### 2) QUALITATIVE EVALUATION

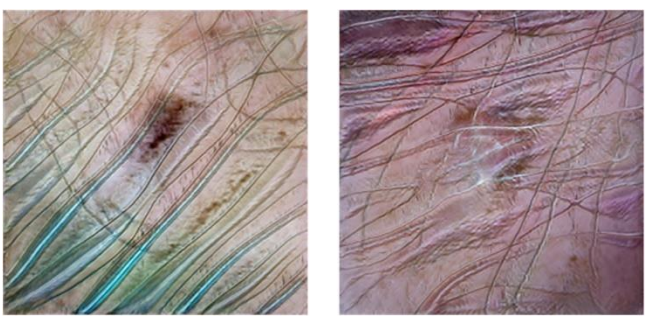


**FIGURE 8. High-frequency ripple artifacts in generated samples of cSG2.**

A side-by-side comparison of generated images produced by cSG2 and STGAN for three minority classes is presented in Fig. 7. The visual inspection clearly highlights a noticeable disparity in the quality of the generated images between cSG2 and STGAN. Notably, images generated by STGAN exhibit a comparable level of fidelity and detail as real



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images, while cSG2 fails to generate some modes of the training images. Furthermore, high-frequency ripple artifacts are observed in the synthetic images generated by cSG2 (Fig. 8). These images lack fine-grained details, which limits the improvement in performance for downstream skin disease diagnostic tasks (subsection IV-F).

### *F. THE PERFORMANCE OF STGAN IN CNN-BASED MULTI-CLASS CLASSIFICATION*

#### 1) CLASSIFICATION RESULTS OF DIFFERENT METHODS

For a comprehensive analysis of STGAN and other key components (T-ResNet50 and TTA) of the proposed image

TABLE III OVERALL CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS

| Method | Accuracy | Sensitivity | Precision | F1-score | Specificity |
|---|---|---|---|---|---|
| ResNet50 | 94.72 | 62.08 | 68.27 | 64.04 | 95.11 |
| T-ResNet50 | 97.76 | 83.81 | 87.42 | 85.43 | 97.91 |
| T-ResNet50 + cSG2 | 97.65 | 83.81 | 88.19 | 85.83 | 97.79 |
| T-ResNet50 + STGAN | 97.86 | 87.18 | 90.29 | 88.62 | 98.06 |
| T-ResNet50 + cSG2 + TTA | 97.86 | 85.98 | 89.65 | 87.64 | 97.93 |
| T-ResNet50 + STGAN + TTA | 98.23 | 88.85 | 90.23 | 89.48 | 98.34 |

TABLE IV COMPARISION RESULTS OF DIFFERENT METHODS ON EACH CLASS

| Metric | Method | MEL | NV | BCC | AKIEC | BKL | DF | VASC | Macro Average |
|---|---|---|---|---|---|---|---|---|---|
| ACC | ResNet50 | 91.46 | 88.32 | 96.51 | 96.61 | 91.46 | 98.90 | 99.80 | 94.72 |
| | T-ResNet50 | 96.41 | 95.46 | 98.55 | 98.25 | 96.16 | 99.70 | 99.80 | 97.76 |
| | T-ResNet50 + cSG2 | 96.06 | 94.96 | 98.55 | 98.55 | 95.86 | 99.60 | 99.95 | 97.65 |
| | T-ResNet50 + STGAN | 95.86 | 95.16 | 98.80 | 98.85 | 96.61 | 99.75 | 100.00 | 97.86 |
| | T-ResNet50 + cSG2 + TTA | 96.31 | 95.16 | 98.85 | 98.55 | 96.51 | 99.70 | 99.95 | 97.86 |
| | T-ResNet50+ STGAN + TTA | 96.61 | 96.06 | 99.10 | 98.60 | 97.55 | 99.75 | 99.95 | 98.23 |
| Sensitivity | ResNet50 | 53.81 | 93.36 | 63.11 | 49.23 | 60.45 | 21.74 | 92.86 | 62.08 |
| | T-ResNet50 | 77.13 | 97.91 | 88.35 | 64.62 | 83.18 | 82.61 | 92.86 | 83.81 |
| | T-ResNet50 + cSG2 | 76.23 | 97.46 | 85.44 | 75.38 | 81.82 | 73.91 | 96.43 | 83.81 |
| | T-ResNet50 + STGAN | 82.06 | 96.87 | 81.55 | 76.92 | 85.91 | 86.96 | 100.00 | 87.18 |
| | T-ResNet50 + cSG2 + TTA | 77.13 | 97.61 | 91.26 | 78.46 | 82.73 | 78.26 | 96.43 | 85.98 |
| | T-ResNet50+ STGAN + TTA | 83.41 | 97.84 | 87.38 | 73.85 | 88.18 | 91.30 | 100.00 | 88.85 |
| Precision | ResNet50 | 63.83 | 89.62 | 67.01 | 47.76 | 61.29 | 55.56 | 92.86 | 68.27 |
| | T-ResNet50 | 89.12 | 95.42 | 84.26 | 77.78 | 82.06 | 90.48 | 92.86 | 87.42 |
| | T-ResNet50 + cSG2 | 86.73 | 95.12 | 86.27 | 79.03 | 80.72 | 89.47 | 100.00 | 88.19 |
| | T-ResNet50 + STGAN | 80.97 | 95.94 | 94.38 | 86.21 | 83.63 | 90.91 | 100.00 | 90.29 |
| | T-ResNet50 + cSG2 + TTA | 88.21 | 95.27 | 87.04 | 77.27 | 85.05 | 94.74 | 100.00 | 89.65 |
| | T-ResNet50+ STGAN + TTA | 85.71 | 96.33 | 94.74 | 81.36 | 89.40 | 87.50 | 96.55 | 90.23 |
| F1-score | ResNet50 | 58.39 | 91.45 | 65.00 | 48.48 | 60.87 | 31.25 | 92.86 | 64.04 |
| | T-ResNet50 | 82.69 | 96.65 | 86.26 | 70.59 | 82.62 | 86.36 | 92.86 | 85.43 |
| | T-ResNet50 + cSG2 | 81.15 | 96.28 | 85.85 | 77.17 | 81.26 | 80.95 | 98.18 | 85.83 |
| | T-ResNet50 + STGAN | 81.51 | 96.40 | 87.50 | 81.30 | 84.75 | 88.89 | 100.00 | 88.62 |
| | T-ResNet50 + cSG2 + TTA | 82.30 | 96.43 | 89.10 | 77.86 | 83.87 | 85.71 | 98.18 | 87.64 |
| | T-ResNet50+ STGAN + TTA | 84.55 | 97.08 | 90.91 | 77.42 | 88.79 | 89.36 | 98.25 | 89.48 |
| Specificity | ResNet50 | 96.18 | 78.10 | 98.32 | 98.19 | 95.29 | 99.80 | 99.90 | 95.11 |
| | T-ResNet50 | 98.82 | 90.48 | 99.11 | 99.38 | 97.76 | 99.90 | 99.90 | 97.91 |
| | T-ResNet50 + cSG2 | 98.54 | 89.88 | 99.26 | 99.33 | 97.59 | 99.90 | 100.00 | 97.79 |
| | T-ResNet50 + STGAN | 97.58 | 91.69 | 99.74 | 99.59 | 97.92 | 99.90 | 100.00 | 98.06 |
| | T-ResNet50 + cSG2 + TTA | 98.71 | 90.18 | 99.26 | 99.23 | 98.21 | 99.95 | 100.00 | 97.93 |



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processing framework for skin lesion diagnosis, a detailed ablation study is provided in Table III.

According to experimental results in Table III, the following observations are made. First, T-ResNet50 outperforms ResNet50 across all five metrics. The underlying reason for this observation is that ResNet50 has over 23 million parameters, making it challenging to effectively fit to a limited dataset like HAM10000,

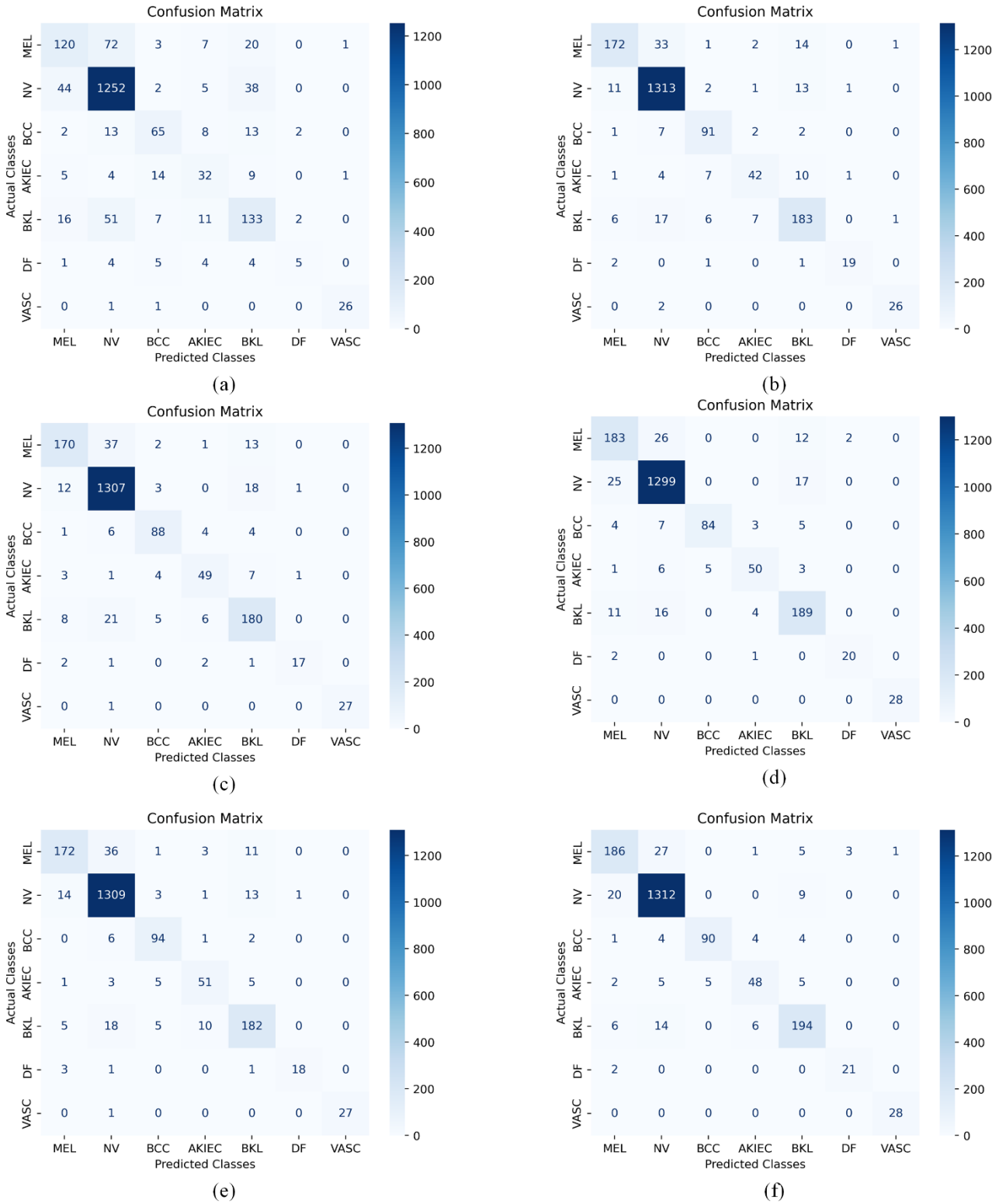



**FIGURE 9. Confusion matrix of the classification methods. (a) ResNet50. (b) T-ResNet50. (c) T-ResNet50 + cSG2. (d) T-ResNet50 + STGAN. (e) T-ResNet50 + cSG2 + TTA. (f) T-ResNet50 + STGAN + TTA.**



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TABLE V Classification Performance (%) Comparison of SOTA Methods on HAM10000 Dataset.

| Reference (Year) | Accuracy | Sensitivity | Precision | F1-Score | Specificity |
|---|---|---|---|---|---|
| Nozdryn-Plotnicki, *et al.* [77] (2018) | 95.8 | 83.3 | 82.6 | 82.3 | **98.6** |
| Gessert, *et al.* [78] (2018) | 97.2 | 80.9 | 88.8 | 84.1 | 98.4 |
| Zhuang, *et al.* [79] (2018) | 96.8 | 80.4 | 86.1 | 83.0 | 98.0 |
| Rashid, *et al.* [27] (2019) | - | 82.86 | 84.43 | 83.44 | - |
| Qin, *et al.* [22] (2020) | 95.2 | 74.3 | 76.9 | 73.2 | 96.6 |
| Hoang, *et al.* [69] (2022) | 95.66 | 70.71 | 75.15 | 72.61 | 96.03 |
| Wei, *et al.* [70] (2023) | - | 83.81 | 83.75 | 83.45 | - |
| Anand, *et al.* [71] (2023) | 97.96 | 84.86 | 88.47 | - | 97.93 |
| T-ResNet50 + STGAN + TTA | **98.23** | **88.85** | **90.23** | **89.48** | 98.34 |

Best results are in boldface.

whereas T-ResNet50 with pre-trained knowledge from ImageNet reduces the demand for the number of training data. Second, STGAN-based augmentation can enhance the classification results of T-ResNet50, and improve all evaluation metrics, particularly in Sensitivity, Precision and F1-score, which is improved by 4%, 3.3% and 3.73%, respectively. On the contrary, cSG2-based augmentation does not show an obvious advantage over T-ResNet50, and even Accuracy and Specificity are worse. Third, TTA can further boost the classification performance of GAN-based augmentation. After applying TTA, the proposed method achieved the highest Accuracy, Sensitivity, F1-score, and Specificity. Compared to T-ResNet50, it improved by 0.48%, 6.02%, 3.21%, 4.74%, and 0.44% in Accuracy, Sensitivity, Precision, F1-score, and Specificity, respectively. Compared to ResNet50, it showed improvements of 3.7%, 43.12%, 32.15%, 39.71%, and 3.4%, respectively.

It is worth noting that the proposed method improves the Sensitivity of both ResNet50 and T-ResNet50 by a large margin, which is crucial in clinical diagnosis of skin diseases, as it indicates a high number of true positives and a low number of false negatives. Consequently, the risk of missing relevant disease cases is minimized, ensuring fewer cases are overlooked during diagnosis.

Further, the performance of different methods on each class is reported in Table IV, and the confusion matrix is presented in Fig. 9. One observation is that the classification result of ResNet50 is heavily affected by uneven distribution of the dataset, as it only correctly classifies 5 out of 23 DF samples and 32 out of 65 AKIEC samples (Fig. 9(a)), leading to poor overall performance on average Sensitivity (62.08%), Precision (68.27%) and F1-score (64.04%). T-ResNet50 is less affected by data imbalance, and significantly improves its recognition capability for these two classes. On the basis of T-ResNet50, by adding STGAN generated images to the training set, the Accuracy, Sensitivity, Precision, and F1-score of AKIEC, BKL, DF, VASC are all improved. Furthermore, with the additional use of TTA, the true postive for MEL, BKL, DF and VASC reach their maximum values leading to the highest average Accuracy, Sensitivity, F1-score and Specificity values among all six methods.

### 2) COMPARISON WITH OTHER WORKS

Table V presents a comparison between the proposed method and recent SOTA works in the field of skin lesion classification. The works in [77], [78], [79] correspond to the top three winners' team of ISIC 2018 Challenge, which use HAM10000 dataset as training set [80], representing a high level of performance in this classification task. The works in [22], [27], [69], [70], [71] correspond to recent representative methods from the literature. It should be noted that, the ISIC 2018 Challenge provided the HAM10000 dataset as training data, with additional testing data not included in the original dataset. Same as this study, the works in [22], [27], [69], [70], [71] also divided the HAM10000 dataset into training and testing sets using an 8:2 ratio.

Although the Specificity of the proposed method is slightly lower than [77], [78], these two works utilized additional data during the training process. An additional proprietary dataset which has 33,644 samples was utilized in in [77] and another public dataset with roughly 13,500 images was utilized in [78]. The proposed method outperforms their Sensitivity, Precision and F1-score by a large margin without using any additional data. The works in [22], [27] also employed GAN-based data augmentation method to mitigate limited data and class imbalance problem. However, [22] did not explore the universal knowledge between different classes, and [27] overlooked the mode collapse problem in conditional GAN arising from data imbalance. This leading to a limited improvement of



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classification performance in [22], [27]. Anand, *et al.* [71] utilized a U-Net model to segment the disease before classification and achieved a high accuracy of 97.96%. However, training the U-Net model needs pixel-level annotations, which entail additional workload in practical applications.

TABLE VI COMPARATIVE ANALYSIS OF SENSITIVITY OF MELANOMA BETWEEN PROPOSED METHOD AND SOTA METHODS.

| Reference (Year) | Sensitivity |
|---|---|
| Nozdryn-Plotnicki, *et al.* [77] (2018) | 76 |
| Gessert, *et al.* [78] (2018) | 80.1 |
| Zhuang, *et al.* [79] (2018) | 70.2 |
| Rashid, *et al.* [27] (2019) | 79 |
| Qin, *et al.* [22] (2020) | 78.5 |
| Hoang, *et al.* [69] (2022) | 64.13 |
| Wei, *et al.* [70] (2023) | 81.45 |
| Anand, *et al.* [71] (2023) | 76 |
| T-ResNet50 + STGAN + TTA | **83.41** |

Best results are in boldface.

Furthermore, since melanoma is the most dangerous and accounts for the highest number of deaths among all skin cancers, improving Sensitivity for melanoma means reduce the risk of malanoma patients being misdiagnosed as other skin diseases. The proposed method is compared to SOTA methods for melanoma Sensitivity in Table VI. This study achieves the highest Sensitivity. This indicates that the proposed method has the potential to contribute to the development of accurate and robust automated skin lesion diagnosis systems, thereby improving the survival chances of melanoma patients.

## V. CONCLUSION AND FUTURE WORK

In this paper, a two-stage GAN-based data augmentation method is proposed to enhance the classification performance of imbalanced skin lesion dataset. The proposed method decouples the learning of universal knowledge and class-specific knowledge to mitigate mode collapse of minority classes. Additionally, self-supervised learning is introduced to improve sample efficiency, and Freeze-D is applied to enhance stability during the learning of class-specific knowledge. Based on the proposed GAN-based data augmentation method, a framework to enhance the classification performance is established. Experimental results on HAM10000 dataset showing that the proposed method can generate diverse and visually-appealing images and can provide additional information to improve the classification results. Accordingly, this study can aid in the development of automated skin disease diagnosis system and contribute to improving the survival rates of individuals with skin cancer. A limitation of this study is that due to the presence of natural artifacts, such as hair and veins, in the original training images, images generated by the proposed method also retain these artifacts. These artifacts may occlude the skin lesions and potentially cause the degradation of the classification performance. As a future direction, we intend to explore the use of image inpainting techniques to remove these natural artifacts before training and further boost the performance of GAN-based data augmentation in CNN-based classification.

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